PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
ARUSHA REGION
FORM TWO MOCK EXAMINATION

Code: 033 **BIOLOGY**

TIME: 2:30 Hours **MAY 2024**

INSTRUCTIONS:

1. This paper consists of sections A, B and C with a total of ten (10) questions.
2. Answer all questions.
3. All writing must be in blue or black ink except drawing which must be in pencil.
4. Cellular phones and any unauthorized materials are not allowed in the assessment room.
5. Write your assessment number at the top of every page.

SECTION A (15 Marks)

1. For the following items choose the correct answer among the given alternative and write its letter in the box provided below:

(i) Immunity is the ability of the body to defend itself against infections. Which one of the following types of immunity involves passage of antibodies from mother to unborn baby across the placenta?

A. Natural passive immunity
B. Artificial active immunity
C. Artificial passive immunity
D. Artificial active immunity
E. Readymade immunity

(ii) A person pick up a telephone after hearing it ring. Which characteristics of living things is the person showing by this action?

A. Movement and locomotion
B. Movement and irritability
C. Growth and response
D. Development and response

(iii) Which apparatus are used for heating substance in the Biology laboratory?

A. Test tube holder and thermometer
B. Spirit lamp and Bunsen burner
C. Bunsen burner and Thermometer
D. Spirit lamp and test tube rack

(iv) Nutritional disorders are common problem associated with poor balanced diet mostly to children. Which among the following is not nutritional Dis-order?

A. Anaemia
B. Goiter
C. Asthma
D. Night blindness

(v) Juma is a carpenter and Agrey is a farmer. A diet taken by Juma and Agrey after their activities should contain large amount of;

A. Carbohydrates
B. Protein
C. Vitamin
D. Fats and oils

(vi) If you are asked to prepare a first aid kit which of the following will not be included?

A. Iodine tincture
B. Medicine to cure disease
C. Pair of scissors
D. Sterilized cotton wool

(vii) What is the main product of anaerobic respiration in animals?

A. Lactic acid
B. Alcohol
C. Water
D. Carbondioxide

(viii) Conditions or factors that can change or be changed in an experiment

A. Hypothesis
B. Problems
C. Variables
D. Conclusions

(ix) Which statement is true about the Virus?

A. It is active outside the host cell but dormant inside it.
B. It is active inside the host cell but dormant outside it.
C. It carries life processes when outside the host cell.
D. It has both DNA and RNA as genetic materials.

(x) _____ Used to catch the flying insects.

A. Fish net
B. Sweep nets
C. Pooter
D. Specimen bottle

(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)	(x)

2. Match the phrase in **LIST A** with responses in **LIST B** by writing the letter of the correct response in a table provided.

LIST A	**LIST B**
(i) Breaking down of food materials within a cell to release energy. (ii) Process by which living things produce new individuals of own kind. (iii)Process of removing metabolic waste products from the body of organism. (iv)Process of taking in food synthesizing it, digesting and oxidizing it to release energy. (v) Ability of an organism to respond to stimulus.	A. Excretion B. Nutrition C. Reproduction D. Respiration E. Irritability F. Growth G. Movement H. Locomotion I. Death

LIST A	i	Ii	iii	iv	v
LIST B					

SECTION B (70 Marks)

Answer **all** questions in this section.

3. (a) Mr. Nguvu asks you to prove that, biology has close relationship with other science fields rather than classroom teaching. (Provide any 3 points)

i. ……

ii. ……

iii. ……

b) Differentiate the following:

(i) Laboratory and Biology laboratory

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(ii) First Aid and First Aid Kit

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4. (a) Explain the following conceptual statement as applied in biology

(i) Vitamin B and C needed to be taken regularly …….

…….

(ii) We should include roughage in our diets even through, it is not a nutrients

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…….

(iii) Ruminant can digest cellulose human being can not …….

……

…….

(iv) A snake bite victim should not drink alcohol…….

……

…….

(v) The ileum is very long about six metres…….

……

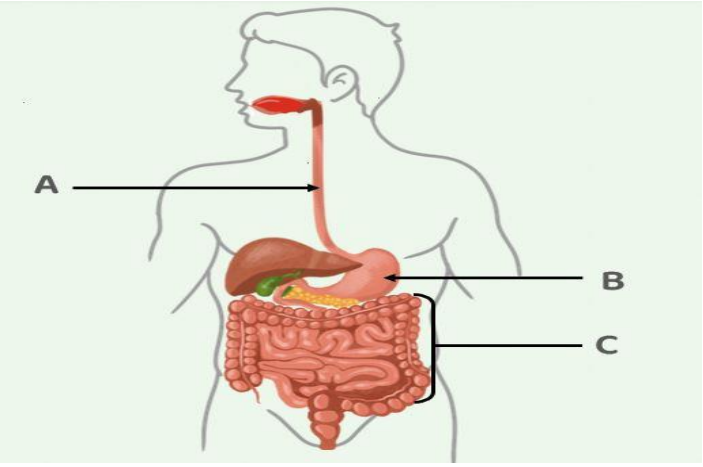
…….

(b) (i) Define alimentary canal

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…….

(ii) Study the diagram below and give the function of parts A, B and C in digestion



5. A group of form II student from Moivo Secondary School had a study visit at St. Francis Hospital in Ifakara. They noted a patient with sunken eyes, severe dehydration, weak pulse, deep breathing, vomiting and watery diarrhea like rice water.

(a) What could be the possible disease that the patient was suffering from?
……

(b) Briefly explain four controls or preventive measures of the disease mentioned in (a) above.

i. ……
ii. ……
iii. ……
iv. ……

6. Assuming that your biology teacher asked you to draw well labeled diagram of animal cell and show the parts responsible for the following:

(i) Protect and give the cell a definite shape.
(ii) Allows selective movement of substance across the cell.
(iii) Site for chemical reaction.
(iv) Home of the hereditary materials.

(b) Highlight differences between Mango tree and cat based on their basic unit of life (two points).

Mango tree	Cat
i.	
ii.	

7 .(a) Define the term personal hygiene;

……

(b) State four principles of personal hygiene

i.

……

……

ii.

……

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iii.

……

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iv.

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……

(c) Give any four **(04)** changes that may occur in both girls and boys at puberty stage.

i. …….

ii. ……

iii. ……

iv. ……..

8. One day **Genel** and **Jayden** went to forest to hunt, unfortunately **Jayden** saw many Bats flying in a big tree, **Jayden** told **Genel** they are birds while they are not.

(c)What type of classification used by **Jayden** on classifying Bats as birds?

…….

(d) Outline three advantages of such kind of classification

i. ……

…….

ii. ……

…….

iii. ……

……

(c) By giving two reasons, why this kind of classification is not good at all?

i. ……

ii. ……

9. (a) It is very difficult to conduct first aid to an accident victim while you are empty handed. Highlight any four (4) requirements items and their uses in providing first aid.

i. ……

ii. ……

iii. ……

iv. ……

(b) Snakes are among dangerous reptiles in the universe. How can you render a first aid to someone who has been bitten by a snake? (Give six (6) points)

i. …….
ii. ……..
iii. …….
iv. ……
v. ……
vi. ……..

SECTION C (15 MARKS)

10. People at Sakina village have bad habit of poorly disposing waste materials in the environment. As biology student educate people of Sakina on the effect of their behavior.